Conservative Endpoint–Bickley Reduction for Low-Froude Michell Resistance

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Abstract

Reliable low-Froude wave-resistance prediction requires both resolving an increasingly oscillatory Michell integral and estimating its algebraic tail. For a conventional real-axis marcher, we found that a quiet-window estimate understated the actual error by a factor of 113. A structural correction makes the estimate conservative over the tested range. For polynomial tensor-product B-spline hulls, the exact inner transform is a finite sum of waves attached to spline-span endpoints. At low Froude number, an explicit absolute bound permits the omission of terms below the waterline. Expanding the squared transform before outer integration then reduces the resistance to pairwise kernels $K_s(\omega) = \int_1^\infty \mathrm{e}^{\mathrm{i}\omega\lambda}\lambda^{-s}(\lambda^2-1)^{-1/2}\,d\lambda$*, which are analytically continued Bickley kernels. The substitution* $\lambda = 1 + t^2$*, followed by the exact rotation* $t = \mathrm{e}^{\mathrm{i}\pi/4}y/\sqrt{|\omega|}$*, converts each nonzero-frequency kernel to a Gaussian-decaying integral whose quadrature cost is independent of frequency. A hybrid implementation accepts the reduced result only when its omission bound and a coarse/fine contour estimate meet the requested tolerance. For the standard Wigley hull at* $Fn = 0.02$*, the method uses 1,152 kernel nodes instead of 511,504 inner-transform evaluations, reduces median runtime from* 3.923×10^{1} ms *to* 4.500×10^{-2} ms *(approximately 880-fold), and differs from an independent real-axis reference by* 1.53×10^{-13} *relatively. The solver is an engineering improvement; its specific combination of known endpoint and contour methods may be new. It applies only to upright symmetric monohulls, and its contour-discretization estimate is not a formal interval certificate.*

KEY WORDS: Wave resistance; Michell integral; low Froude number; oscillatory quadrature; B-splines; Bickley–Naylor function.

Introduction

Michell's 1898 thin-ship theory expresses wave resistance as a one-dimensional integral of the squared Fourier–Laplace transform of the longitudinal hull slope [Michell, 1898, Tuck, 1989]. Compared with nonlinear free-surface methods, Michell's formula costs little and retains the bow–stern interference that couples resistance to speed and hull shape. Designers therefore use it for preliminary design, multihull studies, and mathematical shape optimization [Lazauskas, 2009, Dambrine et al., 2016]. At low speed, however, a fast result is useful only if its error estimate remains trustworthy.

The apparent simplicity conceals a difficult numerical limit. Let $\nu = g/U^2$, where U is speed and g gravitational acceleration. As U and therefore $Fn = U/\sqrt{gL}$ decrease, the phase $\nu\lambda x$ oscillates increasingly rapidly over the unbounded outer coordinate λ. Tuck used Filon-like longitudinal integration, while Lazauskas used exact piecewise-quadratic formulas and a fixed angular rule; Lazauskas reports that very low Froude number remains exceptional [Tuck, 1989, Lazauskas, 2009]. A real-axis adaptive marcher must spend more effort as frequency rises and must decide where to truncate the oscillatory algebraic tail. At the pre-fix revision `64925d4`, one quiet-window criterion reported 5.123×10^{-9} relative error at $Fn = 0.05$, but the measured error was 5.781×10^{-7}, about 113 times larger. The frozen kernel advances quiet windows by oscillatory phase alone and adds the refinement and tail estimates. Across $0.02 \le Fn \le 1$, the corrected estimate covers measured error by about fourfold. In the four low-Froude cases reported below, measured error falls by factors of 3.2–4.5, and evaluations fall by 42–51%.

Earlier work supplies each part of our method. Wehausen [1973] surveys classical thin-ship theory and its limits. Keller and Ahluwalia show that bow and stern waterline data control the small-Froude wave field and resistance [Keller and Ahluwalia, 1976]. Gotman separates polynomial-hull contributions into endpoint derivatives and bow–stern interference terms [Gotman, 2002]. Huybrechs and Vandewalle develop numerical steepest descent [Huybrechs and Vandewalle, 2006], and Motygin applies it to the Kelvin Green-function integral in linear ship-wave theory [Motygin, 2017]. We combine these ideas in four steps:

1. decompose every polynomial B-spline span exactly into finitely many endpoint terms, including terms at interior and full-multiplicity chine knots;
2. bound every resistance pair that contains an omitted submerged endpoint;
3. identify the retained pair kernels as analytically continued Bickley–Naylor functions and evaluate them on a frequency-scaled Gaussian contour; and
4. dispatch conservatively between the reduced solver and the general real-axis method.

Each ingredient has precedent. Our priority claim, therefore, extends only to the combined algorithm and implementation. A targeted search found no source that contains all four elements, but a negative search cannot establish novelty. In particular, we could not obtain the full text of de Sendagorta and Grases [1988], a closely related study.

Michell resistance and spline geometry

Let $y = f(x,z) \geq 0$ denote the half-breadth of an upright symmetric hull, with $z \geq 0$ measured downward from the undisturbed waterline. In the normalization used by the implementation,

$$R_{\mathrm{w}} = \frac{4\rho g^2}{\pi U^2} \int_1^\infty \frac{\lambda^2}{\sqrt{\lambda^2 - 1}} \, |F(\lambda)|^2 \, d\lambda, \tag{1}$$

$$F(\lambda) = \iint_\Omega f_x(x,z) \mathrm{e}^{-\nu\lambda^2 z} \mathrm{e}^{\mathrm{i}\nu\lambda x} \, dx \, dz, \qquad \nu = \frac{g}{U^2}. \tag{2}$$

The hull is a clamped, non-rational tensor-product B-spline surface. On a knot rectangle $S = [x_0, x_1] \times [z_0, z_1]$, its longitudinal derivative is an exact bivariate polynomial,

$$f_x|_S = \sum_{a=0}^{p-1} \sum_{b=0}^{q} c^S_{ab} (x - x_0)^a (z - z_0)^b. \tag{3}$$

The method extracts these coefficients once from exact spline derivatives at span corners; it does not sample stations or waterlines. The implementation discards the zero-length intervals created by repeated knots. An interior knot whose multiplicity equals the degree represents a continuous chine exactly.

Exact endpoint decomposition

The finite polynomial degree makes repeated integration by parts terminate. For a polynomial p of degree m,

$$\int_{x_a}^{x_b} p(x) \mathrm{e}^{\mathrm{i}kx} \, dx = \sum_{r=0}^{m} \frac{(-1)^r}{(\mathrm{i}k)^{r+1}} \left[p^{(r)}(x) \mathrm{e}^{\mathrm{i}kx} \right]_{x_a}^{x_b}, \tag{4}$$

$$\int_{z_a}^{z_b} p(z) \mathrm{e}^{-\kappa z} \, dz = -\sum_{u=0}^{m} \frac{1}{\kappa^{u+1}} \left[p^{(u)}(z) \mathrm{e}^{-\kappa z} \right]_{z_a}^{z_b}. \tag{5}$$

Apply equations (4) and (5) to every monomial in equation (3), with $k = \nu\lambda$ and $\kappa = \nu\lambda^2$. The result is an exact endpoint representation.

Proposition 1 (Finite endpoint representation)**.** *For a polynomial tensor-product B-spline hull with longitudinal degree at least one, the transform in equation (2) can be written*

$$F(\lambda) = \sum_{j=1}^{M} c_j \lambda^{-n_j} \mathrm{e}^{\mathrm{i}\nu\lambda x_j} \mathrm{e}^{-\nu\lambda^2 z_j}, \qquad n_j \geq 3, \tag{6}$$

where (x_j, z_j) are endpoints of nonzero knot spans and the complex coefficients c_j are finite combinations of span-polynomial endpoint derivatives. Equal endpoint/power triples may be combined exactly.

Proof. Equations (4) and (5) replace a monomial of derivative orders (r,u) by endpoint factors proportional to

$$(\nu\lambda)^{-(r+1)} (\nu\lambda^2)^{-(u+1)}.$$

The resulting power of λ is $n = r + 2u + 3$. Both sums terminate at the polynomial degrees, and summing the finite contributions from all nonzero knot rectangles gives equation (6). □

Tests compare equation (6) directly with independent exact-moment calculations for both the Wigley hull and a multi-span hull with a full-multiplicity chine. Across $1 \leq \lambda \leq 20$, every discrepancy falls below the scaled 5×10^{-10} threshold.

Waterline reduction and omission bound

Partition equation (6) into waterline terms W, for which $z_j = 0$, and submerged terms S, for which $z_j > 0$. At low Froude number, every submerged term contains at least $\mathrm{e}^{-\nu z_j}$ because $\lambda \geq 1$. Retaining only W gives

$$\widetilde{F}(\lambda) = \sum_{j \in W} c_j \lambda^{-n_j} \mathrm{e}^{\mathrm{i}\nu\lambda x_j}. \tag{7}$$

Expanding $|\widetilde{F}|^2$ before outer integration removes the nonanalytic complex conjugation from the integrand:

$$\widetilde{R}_{\mathrm{w}} = C \sum_{i,j \in W} \Re\left\{ c_i \overline{c_j} K_{s_{ij}}(\omega_{ij}) \right\}, \qquad C = \frac{4\rho g^2}{\pi U^2}, \tag{8}$$

where $s_{ij} = n_i + n_j - 2 \geq 4$, $\omega_{ij} = \nu(x_i - x_j)$, and

$$K_s(\omega) = \int_1^\infty \frac{\mathrm{e}^{\mathrm{i}\omega\lambda}}{\lambda^s \sqrt{\lambda^2 - 1}} \, d\lambda. \tag{9}$$

Proposition 2 (Submerged-pair error bound)**.** *Let $\mathcal{P}_S = \{(i,j) : i \in S \text{ or } j \in S\}$. The resistance discarded by replacing F with $\widetilde{F}$ satisfies*

$$\left| R_{\mathrm{w}} - \widetilde{R}_{\mathrm{w}} \right| \leq C \sum_{(i,j) \in \mathcal{P}_S} |c_i c_j| \mathrm{e}^{-\nu(z_i + z_j)} K_{s_{ij}}(0). \tag{10}$$

Proof. Expand $|F|^2 - |\widetilde{F}|^2$ into precisely the ordered pairs in $\mathcal{P}_S$. Apply the triangle inequality, use $|\mathrm{e}^{\mathrm{i}\omega\lambda}| = 1$, and bound $\mathrm{e}^{-\nu\lambda^2(z_i + z_j)} \leq \mathrm{e}^{-\nu(z_i + z_j)}$ for $\lambda \geq 1$. The remaining positive integral is $K_{s_{ij}}(0)$. □

The zero-frequency kernel is analytic:

$$K_s(0) = \int_0^\infty \cosh^{-s} t \, dt = \frac{\sqrt{\pi}\, \Gamma(s/2)}{2\Gamma((s+1)/2)}, \tag{11}$$

A stable even/odd recurrence evaluates this kernel.

Bickley kernels and contour quadrature

With $\lambda = \cosh t$, equation (9) becomes

$$K_s(\omega) = \int_0^\infty \mathrm{e}^{\mathrm{i}\omega \cosh t} \cosh^{-s} t \, dt = \mathrm{Ki}_s(-\mathrm{i}\omega), \tag{12}$$

where the final equality denotes analytic continuation of the Bickley–Naylor function [Bickley and Nayler, 1935, NIST Digital Library of Mathematical Functions, 2026].[1] This identity

[1] The 1935 paper is by W. G. Bickley and J. Nayler; later literature and DLMF use the conventional function name "Bickley–Naylor." We retain each spelling in its historical context.

supplies recurrences and other possible implementations. The present solver instead evaluates the contour directly.

The substitution $\lambda = 1 + t^2$ removes the square-root endpoint in equation (9):

$$K_s(\omega) = 2e^{i\omega} \int_0^\infty \frac{e^{i\omega t^2}}{(1+t^2)^s \sqrt{2+t^2}}\, dt. \tag{13}$$

For $\omega > 0$, analyticity in the intervening sector permits the exact rotation $t = e^{i\pi/4} y/\sqrt{\omega}$, giving

$$K_s(\omega) = \frac{2e^{i(\omega+\pi/4)}}{\sqrt{\omega}} \int_0^\infty \frac{e^{-y^2}\, dy}{(1+iy^2/\omega)^s \sqrt{2+iy^2/\omega}}. \tag{14}$$

Conjugation gives the negative-frequency kernels. The rotation replaces oscillation with Gaussian decay and leaves slowly varying frequency dependence in the amplitude. The same numerical-asymptotic principle yields frequency-independent work in high-frequency scattering [Gibbs et al., 2020]; here, the quadratic phase gives the contour in closed form.

The implementation applies 24- and 48-point Gauss–Legendre rules on $0 \le y \le 8$. The neglected Gaussian tail obeys

$$|K_s - K_s^{(8)}| \le \frac{e^{-64}}{8\sqrt{2\omega}}, \qquad \omega > 0, \tag{15}$$

using $|1 + iy^2/\omega| \ge 1$, $|\sqrt{2 + iy^2/\omega}| \ge \sqrt{2}$, and the standard Gaussian-tail bound. At the minimum supported nonzero frequency $\omega = 25$, this is below 4×10^{-30}. The 24/48 difference estimates discretization error. This difference remains empirical. Thus the submerged-term bound and analytic contour-tail bound are rigorous, but the total floating-point error is not certified.

For fixed geometry, every nonzero endpoint pair costs 72 scalar nodes, independent of $\nu|x_i - x_j|$; self pairs use equation (11) without quadrature. The biquadratic Wigley representation has one longitudinal span and one vertical span. Its 16 endpoint terms include eight at the waterline, which form 16 distinct nonzero-frequency bow–stern pairs. Each pair uses $24 + 48 = 72$ contour nodes, for $16 \times 72 = 1{,}152$ scalar nodes in all; equal-position pairs use equation (11). The committed harness checks each count.

Hybrid algorithm and scope

The reduced method supplements the general solver. The following gates choose between them:

1. validate the physical conditions and exact spline geometry;
2. form and combine the exact endpoint terms in equation (6);
3. reject the specialization unless the hull is upright, symmetric, has its spline domain beginning exactly at $z = 0$, and has longitudinal degree at least one;
4. reject it if any distinct retained endpoint pair has $0 < |\omega_{ij}| < 25$;
5. evaluate equation (8), the bound equation (10), and the weighted 24/48 contour difference; and
6. use the reduced result if their combined relative estimate meets the requested tolerance; otherwise run the general real-axis marcher.

The frequency gate reserves the fixed contour for sufficiently rapid oscillation. For the Wigley bow–stern pair, $\omega = \nu L = gL/U^2 = 1/Fn^2$, which equals the implementation's $\nu|x_i - x_j|$. Thus $\omega \ge 25$ corresponds to $Fn \le 0.2$. The error gate then bases the final choice on the actual hull, speed, and resistance cancellation rather than on nominal Froude number alone. At the default relative tolerance 10^{-5}, it rejects the Wigley reduction at $Fn = 0.08$ and accepts it at $Fn = 0.05$.

The implementation uses safe Rust and no external numerical library. Tests check the reduced solver at three independent levels: exact inner decomposition, isolated Bickley kernels, and total resistance. The broader validation harness checks Froude similarity, zero camber for symmetric sides, classical catamaran $4\cos^2$ interference, tolerance self-convergence, and reported against observed error.

Numerical experiments

Independent reference

The standard Wigley half-breadth is

$$f(x,z) = \frac{B}{2}\left(1 - \frac{4x^2}{L^2}\right)\left(1 - \frac{z^2}{T^2}\right), \quad |x| \le L/2, \quad 0 \le z \le T, \tag{16}$$

with $L = 1.000 \times 10^1$ m, $B = 1.000$ m, and $T = 6.250 \times 10^{-1}$ m. The independent reference shares neither the production moment code nor its outer integrator. Instead, it evaluates the analytic Wigley transform on the positive real λ axis with 16-point Gauss–Legendre panels and Kahan-compensated accumulation. Panel boundaries advance the bow phase by at most $\pi/2$, and integration continues to $\lambda = 4000$. At $Fn = 0.02$, doubling this cutoff to 8000 changes the reference by 1.23×10^{-15} relatively. Because floating-point effects remain, this check measures resolution but does not provide an interval bound.

As a published-value anchor, Doctors and Beck [1987, Table 1] report the classical thin-ship value $10^3 C_w = 1.2486$ for this Wigley hull at $Fn = 0.35$. The committed harness gives $10^3 C_w = 1.247922$, a relative difference of 5.43×10^{-4} (0.0543%). We classify this agreement as a known result reproduced (class 1).

Table 1 shows two regimes. At $Fn = 0.08$, submerged terms remain material, and the estimate rejects the reduction at default tolerance. At $Fn \le 0.05$, reduced/reference differences range from 1.53×10^{-13} to 1.03×10^{-12} and remain within the reduced solver's empirical estimates. The general marcher's error, by contrast, grows as Froude number falls. At $Fn = 0.05$, its corrected estimate is 5.774×10^{-7}, versus 1.493×10^{-7} measured. The estimate therefore covers the observed error. This independently checked marcher provides every comparison in table 1.

The isolated kernel tests compare equation (14) against dense real-axis integration for $s \in \{4, 7, 10\}$ and $|\omega| \in \{25, 100, 400\}$, with scaled discrepancy below 10^{-9}. All 215 Rust tests pass, in-

Table 1: Resistance validation against the independent analytic-Wigley, resolved-real-axis reference. "Reduced estimate" combines the rigorous submerged-pair bound and empirical contour difference. The marcher uses requested rel_tol = 10^{-8} and at most six refinements.

Fn	Reference R_{w} (N)	Reduced rel. diff.	Reduced estimate	Marcher rel. diff.
0.08	1.946×10^{-1}	1.172×10^{-5}	3.814×10^{-5}	6.918×10^{-8}
0.05	1.058×10^{-2}	2.070×10^{-13}	3.728×10^{-12}	1.493×10^{-7}
0.03	5.114×10^{-4}	1.034×10^{-12}	3.462×10^{-12}	3.322×10^{-7}
0.02	4.417×10^{-5}	1.526×10^{-13}	7.620×10^{-13}	6.334×10^{-7}

cluding one documentation test; all 11 Python-interface tests also pass.

Runtime

Timing used an optimized build, 30 warmed samples, and the default relative tolerance 10^{-5}. Absolute timings are host-dependent; the work counts and checksums are deterministic. One benchmark process measured both $Fn = 0.02$ methods.

At $Fn = 0.02$, the endpoint solver is approximately 880 times faster. Its cost changes little between $Fn = 0.05$ and 0.02, as equation (14) predicts. During the 21-speed design sweep, the hybrid selects the path that satisfies the error gate. The checksum is $2.553251156101 \times 10^5$; it differs from the pre-fix value because the corrected marcher retains a positive tail that the earlier rule truncated.

Relation to prior work and scope of novelty

The endpoint structure is well established. Keller and Ahluwalia [1976] express leading small-Froude resistance and waves through bow and stern waterline slopes. Gotman [2002] repeatedly differentiates polynomial hull functions, separates endpoint self terms from bow–stern interference, and discusses higher endpoint derivatives. Our finite B-spline decomposition puts that known mechanism into a form suitable for implementation; it does not introduce a new physical asymptotic principle.

Tuck and Lazauskas also establish exact or Filon-like integration of piecewise-polynomial inner data [Tuck, 1989, Lazauskas, 2009]. As Lazauskas [2009] describes it, Michlet combines exact piecewise-quadratic longitudinal formulas with a fixed angular rule. Our solver retains the same thin-ship physics but, for supported B-spline hulls, makes low-Froude outer work independent of frequency. We classify that result as an engineering improvement and make no head-to-head performance claim against the Michlet executable. Exact endpoint reduction also survives the outer modulus square: pair expansion converts it into a small set of reusable special-function kernels. We found no use of the relation $K_s = \mathrm{Ki}_s(-\mathrm{i}\omega)$ in the Michell sources we inspected. Ruffa and Toni give finite Bessel–Struve module representations for integer-order Bickley functions [Ruffa and Toni, 2026]; their stability on the imaginary axis remains to be tested as a possible kernel backend.

Contour deformation is well established for oscillatory quadrature [Huybrechs and Vandewalle, 2006]. Motygin [2017] applies steepest descent and Clenshaw–Curtis quadrature in ship-wave theory, but to the oscillatory part of the Kelvin Green function rather than Michell's resistance integral. Automated methods now handle general phases and coalescing saddles [Gibbs et al., 2024]. Such methods could treat multihull transverse phases; the present quadratic phase needs only the closed contour in equation (14).

Our literature search included publisher and DOI records, arXiv, TRID, the Adelaide repository, DLMF, and backward references in the sources above. It did not find a previous method combining arbitrary-degree B-spline endpoint terms, Bickley analytic continuation, frequency-scaled Gaussian quadrature, and a submerged-endpoint error gate for Michell resistance. The closest unresolved JSR source is de Sendagorta and Grases [1988]: we verified its bibliographic record and abstract but could not obtain the full text, so we cannot determine whether it contains a finite endpoint-pair expansion or a related low-speed construction. The search was not exhaustive and included no patent search. We therefore classify the algorithmic combination as possibly new, subject to full-text and expert literature review.

Limitations and open extensions

The evidence covers a narrow model: upright symmetric monohulls in linear, deep-water thin-ship theory.

- **Geometry.** The reduced solver supports upright symmetric single hulls. Asymmetric, heeled, and multihull paths fall back to the general method.
- **Error certification.** The submerged-pair omission bound and Gaussian tail bound are analytic. The 24/48 quadrature difference and floating-point roundoff are not interval-certified.
- **Reference floor.** The independent real-axis reference uses compensated summation through $\lambda = 4000$; at $Fn = 0.02$, a cutoff-doubling check changes it by 1.23×10^{-15}. Differences near the observed 10^{-13} scale are resolution measurements, not interval-certified digits.
- **Physical model.** Michell theory is linear, inviscid, slender, deep-water, and fixed-attitude. At very low speed, viscous effects may dominate the already small wave resistance. For free-surface flow past singular obstacles, low-Froude waves can be exponentially small, beyond all algebraic orders, and controlled by Stokes phenomena; that different problem is not resolved by better quadrature [Trinh and Chapman, 2015].

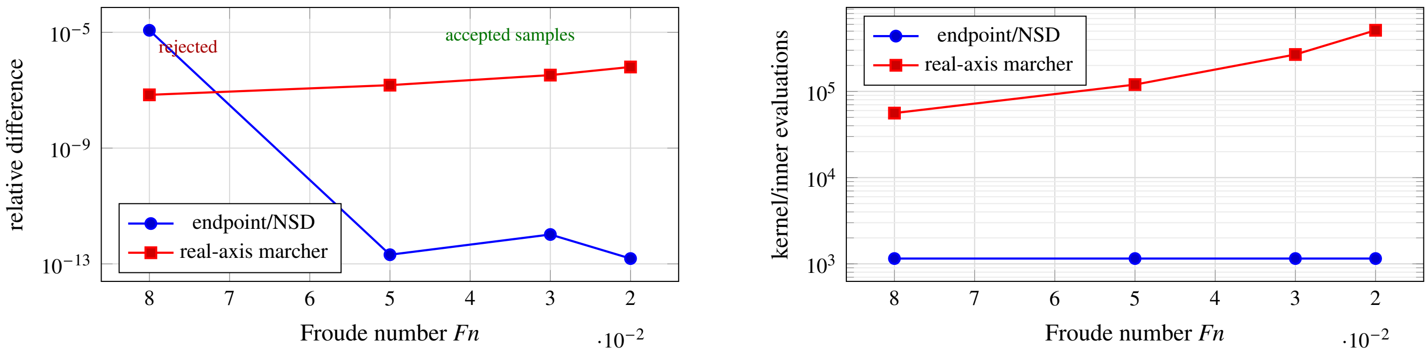

Figure 1: Observed accuracy (left) and work (right) as Froude number decreases. The independent reference resolves endpoint/NSD differences at the 10^{-13}–10^{-12} level, while geometry fixes its quadrature work. At default tolerance, the hybrid rejects the reduced result at $Fn = 0.08$ and accepts it at $Fn = 0.05, 0.03, 0.02$.

Table 2: Final release benchmark on the development host.

Case	Median (ms)	Best (ms)	Diagnostic
21 speeds, $Fn = 0.10 \ldots 0.50$	26.289	22.843	corrected checksum
Default API, $Fn = 0.05$	0.050	0.050	1,152 nodes
Real-axis marcher, $Fn = 0.02$	39.233	35.950	511,504 evaluations
Endpoint/NSD, $Fn = 0.02$	0.045	0.044	1,152 nodes

The highest-leverage numerical extension is multihull support. A transverse placement introduces phase $\nu y \lambda \sqrt{\lambda^2 - 1}$ and moving saddles that equation (14) does not contain. A uniform or automated contour must account for them. Retaining shallow submerged endpoints likewise produces mixed linear and quadratic complex phases. Further work should certify the contour error, test imaginary-argument Bickley recurrences, derive finite-depth kernels, and differentiate the endpoint representation for low-Froude hull optimization.

Conclusions

Finite integration by parts converts the B-spline inner transform into endpoint waves. Pair expansion then turns the outer modulus square into continued Bickley kernels, and contour rotation replaces real-axis oscillation with Gaussian decay. Geometry, rather than Froude number, sets the reduced quadrature work. The submerged-pair bound lets the hybrid reject the reduction whenever it exceeds the requested tolerance.

For the standard Wigley hull at $Fn = 0.02$, the reduced solver is more accurate than the general marcher and approximately 880 times faster. This result establishes an engineering improvement for the supported geometry. Our literature search suggests that the specific combination may be new, but that classification still requires full-text and expert review. Claims beyond the current scope also require broader geometry and certified total error control.

Reproducibility statement

The implementation, tests, benchmark harness, generated results, and number audit are available in the public `michell` repository. The immutable version supporting this paper is archived at `10.5281/zenodo.REPLACE-ME`. The numerical kernel is frozen at repository revision `30d8f53`.

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